

Adam Sultan

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

B.S. in Information Systems & Human-Computer Interaction; GPA: 3.56/4.0

Expected May 2027

Relevant Coursework: Data Structures & Algorithms, Computer Systems, Software Engineering Foundations, Software Construction, Database Design & Development, Application Design & Development, AI & Machine Learning

EXPERIENCE

Allegheny County DHS (Analytics, Technology, & Planning)

Pittsburgh, PA

Software Engineering Intern

Feb 2026 – Present

- Engineering a full-stack “Report Issue” module using **React**, **CSS** and **Django**, integrating with **Jira** to generate structured tickets for team triage.
- Integrated **PostHog** for analytics/observability and contributed via **GitHub** PRs and code review, helping the team track reporting trends and maintain stability in a large shared codebase.

The Silicon Project

New York, NY

Board Director & Lead Developer

Jun 2022 – Present

- Co-architected the nonprofit’s platform using **Next.js** and **Supabase (Postgres)**; built an internal employee portal and a **Stripe-powered** e-commerce store.
- Developed secure backend systems to track donor data and inventory logistics, enabling the processing of **\$430K+** in donations and **12,000+** lbs of e-waste.
- Established engineering standards including **GitHub** branching strategies, PR review workflows, and reusable component libraries to scale volunteer contributions.

Intelliminds

New York, NY

Founding Software Engineer

Aug 2023 – Feb 2025

- Engineered a custom CRM web application using **Next.js**, **TypeScript**, and **Supabase** to automate client-tutor scheduling, reducing manual coordination time by **50%**.
- Implemented SEO optimization and analytics tracking for the public-facing site, driving **1,000+** unique visits and increasing client acquisition.
- Led a team of developers through **Agile sprints**, managing the full SDLC from feature definition to deployment.

PROJECTS

NodeBB Open-Source Contributor | *JavaScript/Node.js, Redis, GitHub Actions, Stryker*

Sep 2025 – Nov 2025

- Developed a “Post Anonymously” plugin in a team fork; implemented server/client hooks, admin settings UI, and permission checks (16 PRs merged).
- Configured **CI/CD pipelines** with GitHub Actions and introduced Stryker mutation testing to harden edge cases and ensure upstream API compatibility.

Concurrent HTTP Proxy Server | *C, Sockets, Pthreads, HTTP, Synchronization*

Nov 2025

- Built a concurrent HTTP/1.0 proxy in C using POSIX sockets and **pthread**s; handled multiple clients with robust request parsing/forwarding and **Rio** buffered I/O.
- Implemented a thread-per-connection model with detached threads and defensive error handling to prevent resource leaks and improve reliability under concurrent load.

Dynamic Memory Allocator | *C, Git, Valgrind, GDB*

Oct - Nov 2025

- Built a 64-bit memory allocator in C (**malloc/free/realloc/calloc**) with **16-byte alignment** and heap growth via **mem_sbrk**.
- Reduced fragmentation with block **splitting/coalescing**; debugged and verified heap invariants using a **heap checker**, **mdriver-dbg**, **GDB**, and **Valgrind**.

TECHNICAL SKILLS

Languages: Java, C, Python, TypeScript, JavaScript, SQL (PostgreSQL), HTML/CSS, R, Ruby

Frameworks: React, Next.js, Node.js, Tailwind CSS, Django, Ruby on Rails

Developer Tools: Git/GitHub, Docker, Linux/Unix, Make, GDB, Valgrind, Figma, Vercel, Supabase

Concepts: Systems Programming, Agile/Scrum, CI/CD, MVC Architecture, Database Design